Project: SURA 2026

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Executive Summary

This report summarizes progress made in Weeks 1–2 of the SURA project and outlines the detailed work plan for Weeks 3–4. The foundational theoretical and computational framework from Horiguchi & Kobayashi (PRX Life, 2025) has been fully studied, implemented, and validated. We are now ready to transition to the bacterial population model - the core biological application of the project.

Work Completed (Weeks 1–2)

Theoretical Study - PRX Life Paper (Horiguchi & Kobayashi, 2025)

Complete study of the optimal control framework for stochastic reaction networks with KL-divergence cost. Topics mastered:

- **Stochastic reaction network formulation** (CME, counting process representation)
- **Hamilton–Jacobi–Bellman (HJB) equation** via dynamic programming
- **KL divergence as control cost** - why it linearizes the HJB
- **Cole–Hopf substitution** Z = exp(V) → linear system
- **Feynman–Kac probabilistic representation**: Z(n,β) = E_k0[exp(β·T_exit)]
- **Martingale proof** of Feynman–Kac formula
- **β and βc** (critical control strength) - physical meaning
- **Mode-switching phenomenon** (ON/OFF phases) in optimal control
- **Legendre conjugate** E(n,u) = sup_β[β·u − V(n,β)]

Computational Implementation

Full pipeline implemented in Python and JavaScript:

- **Gillespie SSA** from scratch (exact stochastic simulation algorithm)
- **O(N) Thomas algorithm** (tridiagonal solver) for linearized HJB
- **Optimal rate computation**: k†_r = k^0_r · Z(n±1)/Z(n)
- **βc binary search** via Thomas solver singularity detection
- **Feynman–Kac Monte Carlo validation** (log-sum-exp stable implementation)
- **E(n,u) Legendre conjugate** via finite difference ∂V/∂β (exact, no MC needed)

Benchmark Reproduction - PRX Paper Figures

All key figures from the paper reproduced computationally:

- **Fig 4(d):** Value function $V(nA, \beta)$ for $\gamma = 1.0, 1.25, 1.5, 1.75$, $N = 100$
- **Fig 4(e):** Gradient ∇V and mode-switching regions
- **Fig 4(f):** Optimally controlled SSA trajectories vs uncontrolled
- **E(n,u) Legendre conjugate curves** (Fig 4b,c) - analytical computation
- **V vs β divergence** at βc

Interactive Visualization Tool

Built and deployed a full interactive dashboard at https://prx-ruddy.vercel.app/ featuring:

- **8 graph types**: V(nA), Z(nA), ∇V mode-switching, optimal rates k†, SSA trajectories, V vs β divergence, E(n,u) Legendre conjugate, Feynman–Kac MC validation
- **Real-time parameter control**: N (up to 5000), γ, k_2, β/βc, n_0
- **β animation** showing value function evolution
- **Multi-β and multi-γ overlays** matching paper's figures
- **Feynman–Kac Monte Carlo validation** confirming $Z(n,\beta) = E[\exp(\beta \cdot T\_exit)]$
- **Computed values panel**: βc, k†, n★, OFF-mode states, E(n,u)

Week 3–4 Work Plan: Bacterial Model

Biological Model - Two-Species Reaction Network

Extend from abstract Moran process to clinically relevant bacterial model with five reactions:

- **R1:** S → 2S with rate k_1 (sensitive bacteria birth, fixed biology)
- **R2:** S → ∅ with rate $k_2(t)$ (antibiotic killing - TIME-VARYING, our control variable)
- **R3:** R → 2R with rate k_3 (resistant bacteria birth, fixed biology)
- **R4:** R → ∅ with rate k_4 (resistant bacteria natural death, fixed biology)
- **R5:** S → R with rate k_5 (mutation: sensitive becomes resistant, fixed biology)

State: $(nS, nR) \in \mathbb{Z}^2{\geq}0$ - 2D instead of 1D for Moran

Pharmacokinetic Profile for $k_2(t)$

Model realistic drug concentration dynamics:

- $C(t) = C_0 \cdot \exp(-\lambda \cdot t\_since\_dose)$ - first-order elimination kinetics
- $k_2(t) = kmax \cdot C(t) / (C(t) + EC_{50})$ - Hill equation (saturating effect)
- Parameters: C_0 (peak dose), λ (clearance rate), EC_{50} (half-maximal concentration), kmax
- Dose every Δt hours → sawtooth concentration profile
- This is the critical extension beyond Horiguchi & Kobayashi: rates are now **time-varying**

Modified Gillespie SSA for Two-Species System

Extend SSA to handle 5 reactions with 2D state and time-varying rates:

- Compute all 5 propensities at each step: $a_1=k_1nS$, $a_2=k_2(t)nS$, $a_3=k_3nR$, $a_4=k_4nR$, $a_5=k_5nS$
- Total rate: $a_0 = a_1+a_2+a_3+a_4+a_5$
- Waiting time: $\tau \sim \text{Exp}(a_0)$
- Reaction selection: fire reaction r with probability ar/a_0
- Update $k_2(t)$ at each step using current time and PK profile
- Track both nS and nR simultaneously

Baseline Characterization (Uncontrolled System)

Before applying optimal control, characterize the natural dynamics:

- Run 1000+ SSA trajectories with standard fixed dosing.
- Measure: average time to eradication, P(resistance emerges), distribution of nR peaks.
- Compare: frequent small doses vs infrequent large doses vs continuous infusion.
- Identify: under what conditions does resistance establish? (nR > threshold)

Optimal Control for Bacterial Model

Extend the HJB/Cole–Hopf framework to the 2-species system:

- **Value function:** V(nS, nR, t) - now 2D state space
- **Cole–Hopf:** Z(nS, nR, t) = exp(V(nS, nR, t))
- **Optimal killing rate:** $k\dagger_2(nS, nR, t) = k^2{}_0(t) \cdot Z(nS-1, nR, t) / Z(nS, nR, t)$
- All other optimal rates similarly derived from Z ratios.
- For small populations: solve 2D linear system (matrix method).
- For large populations: Feynman–Kac Monte Carlo (scalable).

Deliverables for End of Week 4

- Working two-species SSA with pharmacokinetic $k_2(t)$
- Baseline resistance emergence probability under standard dosing
- First version of optimal dosing schedule $k\dagger_2(t)$ from extended framework
- Comparison: controlled vs uncontrolled trajectories
- Initial Pareto frontier: drug cost vs resistance probability
- All code documented and reproducible